

PALESTINE 6 WNW, TX, USA

WMO: 747490

Lat: 31.7796N

Lon: 95.7232W

Elev: 117

StdP: 99.93

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 53968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-1.7	-11.6	1.4	-0.3	-9.0	1.8	2.5	5.8	8.5	5.4	8.2	1.3	N/A	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	37.1	23.2	35.9	23.4	34.7	23.5	25.7	31.7	25.2	31.4	24.9	30.9	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.3	27.7	23.8	18.9	27.5	23.4	18.4	27.3	79.3	32.0	77.6	31.6	76.0	30.9	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.0	3.7		-7.9	38.8	2.3	1.6	-9.5	39.9	-10.8	40.8	-12.1	41.7	-13.7	42.8	
			DB	-9.1	26.3	2.6	0.6	-11.0	26.7	-12.5	27.0	-13.9	27.3	-15.8	27.7	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	9.3	11.4	15.7	19.0	23.1	27.0	28.4	28.8	25.9	20.2	14.5	10.2
	DBStd	8.10	5.59	5.88	5.08	4.16	3.36	2.08	1.78	2.12	2.82	4.40	5.38	5.44
	HDD10.0	231	81	49	13	2	0	0	0	0	0	1	19	66
	HDD18.3	1069	283	203	109	42	6	0	0	0	0	32	137	257
	CDD10.0	3701	60	88	192	271	405	511	569	584	477	318	154	73
	CDD18.3	1498	4	8	29	62	153	261	311	326	227	91	22	6
	CDH23.3	15117	12	38	147	423	1271	2727	3513	3887	2237	742	101	20
	CDH26.7	6858	0	3	16	78	416	1248	1764	2072	1027	223	11	1
(14) Wind	WSAvg	1.7	1.8	2.1	2.1	1.9	1.7	1.4	1.3	1.4	1.4	1.5	1.6	1.7
(15) Precipitation	PrecAvg	1187	103	101	97	94	98	109	68	85	94	114	97	107
	PrecMax	1662	208	229	181	221	199	252	275	245	194	315	319	232
	PrecMin	676	9	3	24	19	0	0	0	0	37	0	5	14
	PrecStd	270	55	59	42	51	60	69	59	68	46	86	69	61
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	26.8	28.4	31.4	33.7	37.3	38.2	39.8	37.4	32.6	28.2	25.6
		MCWB	16.5	17.1	18.0	21.5	22.8	23.5	23.2	23.0	21.8	21.3	20.5	18.6
	2%	DB	22.7	24.4	26.6	28.7	31.9	35.1	36.6	37.6	35.2	30.8	26.0	23.3
		MCWB	15.8	17.3	18.0	20.4	22.4	23.4	23.3	23.2	22.4	21.5	19.6	18.4
	5%	DB	20.7	22.3	25.0	27.2	30.5	33.8	35.3	36.2	33.6	29.3	24.2	21.3
		MCWB	15.7	16.9	17.6	19.6	22.3	23.8	23.7	23.6	22.6	20.7	18.4	17.3
	10%	DB	18.8	20.4	23.4	25.9	29.1	32.5	33.9	35.0	32.1	27.6	22.6	19.7
		MCWB	14.3	16.0	16.8	18.9	21.8	24.0	24.0	23.6	22.2	19.8	18.3	16.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	20.0	20.5	23.0	24.5	26.1	26.2	26.3	25.3	24.7	22.1	20.8
		MCDB	22.3	23.6	24.7	27.6	29.8	31.8	31.8	32.8	30.6	28.2	25.9	23.1
	2%	WB	17.8	19.1	19.8	21.9	23.9	25.5	25.6	25.6	24.7	23.6	20.9	19.5
		MCDB	21.0	22.5	23.9	26.3	29.3	31.3	31.6	32.2	30.1	27.7	24.1	22.4
	5%	WB	16.4	18.2	19.1	21.2	23.3	25.1	25.2	25.1	24.2	22.7	20.2	18.2
		MCDB	19.1	21.2	23.0	25.7	28.7	30.8	31.3	31.7	29.6	26.9	23.1	21.0
	10%	WB	14.4	16.8	18.3	20.5	22.8	24.6	24.8	24.7	23.8	21.9	18.9	16.4
		MCDB	17.2	20.0	22.1	24.9	27.8	30.1	30.8	31.1	29.2	25.6	22.2	19.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	10.5	11.1	11.5	10.4	10.1	10.8	11.4	11.3	11.9	11.4	11.0
		MCDBR	12.6	11.7	11.2	11.7	11.3	11.9	12.8	13.2	12.8	12.3	11.3	12.0
		MCWBR	7.9	6.7	5.9	5.4	4.4	3.2	2.9	2.9	3.7	5.0	6.3	7.9
	5% WB	MCDBR	9.7	9.2	9.6	9.8	9.8	9.7	10.4	11.0	10.6	9.8	9.5	10.3
		MCWBR	7.7	6.8	6.2	5.4	4.4	3.6	3.3	3.0	3.5	5.0	6.5	8.3
(40) Clear-Sky Solar Irradiance	taub		0.320	0.341	0.364	0.406	0.441	0.449	0.457	0.459	0.416	0.365	0.354	0.328
	taud		2.511	2.462	2.407	2.302	2.256	2.273	2.288	2.273	2.345	2.470	2.450	2.498
	Ebn at Noon		906	917	917	887	854	841	833	828	852	879	857	876
	Edh at Noon		87	100	113	131	138	136	133	133	119	98	92	84
(44) All-Sky Solar Radiation	RadAvg		2.82	3.17	4.34	5.37	5.79	6.38	6.49	6.08	5.12	4.14	3.02	2.46
	RadStd		0.26	0.56	0.40	0.36	0.44	0.54	0.45	0.43	0.45	0.58	0.38	0.32

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+88

Nomenclature: See separate page